

Data Centers & Water

in the Edwards Aquifer Region

A 7-county screening of estimated data-center water use, recharge-zone exposure, and growth.

28

facilities catalogued

~15,928

ac-ft/yr est. water
conservative floor

7

county study area

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Table of Content

Abbreviations and nomenclature	3
Abbreviations and units	3
Nomenclature and key terms	3
1. Executive summary	5
2. Study area and scope	5
3. Data collection	6
3.1 Data sources used in this screening	6
3.2 Datasets planned for the full study (not used in this preliminary screening)	6
4. Methodology and calculations	7
4.1 Constants and conversion factors	7
4.2 Water-use model	7
4.3 Derivation of the uniform water factor	8
4.4 Worked per-facility examples	8
4.5 Aggregation	8
4.6 Recharge-zone exposure	8
4.7 Household-equivalent context	8
4.8 Growth and historical rate	8
4.9 Projection to 2050	9
4.10 Load-factor sensitivity	9
5. Results	9
5.1 Totals (quantified sites)	9
5.2 By county	9
5.3 By operator	9
5.4 Illustrative 2050 projection scenarios	10
6. Data quality, assumptions, and limitations	10
7. Capacity research — findings on unquantified sites	11
8. Policy findings and options	11
9. References and data sources	12
Appendix A — Full facility inventory (28 sites)	13

Abbreviations and nomenclature

Abbreviations, units, and key terms used throughout this report.

Abbreviations and units

Abbreviation	Meaning
ac-ft	Acre-foot (1 ac-ft = 1,233.48 m ³)
CAGR	Compound annual growth rate
cfs	Cubic feet per second (springflow)
DC	Data center
EAA	Edwards Aquifer Authority
ET	Evapotranspiration
GCD	Groundwater conservation district
GIS	Geographic information system
GRACE	Gravity Recovery and Climate Experiment (NASA)
GWL	Groundwater level
kWh	Kilowatt-hour
L/kWh	Liters per kilowatt-hour (water-use efficiency unit)
LULC	Land use / land cover
MODIS	Moderate Resolution Imaging Spectroradiometer (NASA)
MW	Megawatt
m ³	Cubic meter
TAC	Texas Administrative Code
TCEQ	Texas Commission on Environmental Quality
TDLR	Texas Department of Licensing and Regulation
TWDB	Texas Water Development Board
TxDOT	Texas Department of Transportation
USGS	U.S. Geological Survey
WUE	Water-use efficiency (L/kWh)

Nomenclature and key terms

Term / symbol	Definition
Quantified site	Facility with a known capacity; included in the totals
Unquantified site	Facility lacking capacity; excluded from totals (conservative)

Term / symbol	Definition
Recharge zone	Area where surface water enters the aquifer
Contributing zone	Upgradient catchment that drains toward the recharge zone
Load factor	Assumed fraction of capacity at which a facility operates
Household-equivalent	Water normalized at 0.4 ac-ft per household·yr (context only)
Point-in-polygon	Geometric test for whether a point lies inside a boundary
r	Assumed annual growth rate in the 2050 projection
W(t)	Estimated water use in year t

1. Executive summary

This report screens the estimated water footprint of data centers across the seven-county Edwards Aquifer study area (Bexar, Comal, Hays, Kinney, Medina, Travis, and Uvalde). It documents the data collected, the full calculation chain behind every figure, and the resulting estimates, together with an explicit account of assumptions and limitations.

Headline estimate. Of 28 catalogued facilities, 14 have a known capacity and are quantified here. These 14 sites carry an estimated **15,928 acre-feet per year** of water use, on **1,246 MW** of capacity. Because the other 14 sites lack capacity data, this total is a conservative floor and understates the true regional figure.

- **Concentration:** water use is dominated by Bexar (7,747 ac-ft/yr) and Medina (4,474 ac-ft/yr).
- **Recharge exposure:** 6 of 14 quantified sites fall on the (approximate) Edwards recharge zone, representing ~5,753 ac-ft/yr — roughly 36% of the quantified footprint.
- **Growth:** the catalogued footprint grew at about 13.2% per year over 2017–2027; an illustrative projection to 2050 is highly sensitive to the assumed rate (see Section 5).
- **Scope notes:** data centers appear in only 5 of the 7 counties (none catalogued in Uvalde or Kinney). Samsung Austin is a semiconductor fab and is reported separately, not in the data-center total.

All water figures are engineering estimates derived from capacity and a water-use-efficiency coefficient (Section 4), not metered withdrawals.

2. Study area and scope

The study area is the seven-county Edwards Aquifer region of south-central Texas. The aquifer's recharge and contributing zones trend southwest-to-northeast along the Balcones Escarpment through these counties, making facility siting relative to the recharge zone a central concern.

Coverage. Catalogued data centers occur in 5 of the 7 counties; quantified (capacity-known) sites occur in 4. Uvalde and Kinney contain no catalogued facilities. The table below summarizes county coverage among quantified sites.

County	Quantified sites	Capacity (MW)	Est. water (ac-ft/yr)
Bexar	7	606	7,747
Medina	2	350	4,474
Hays	1	150	1,918
Travis	4	140	1,790
Comal	0 (1 unquantified)	—	—
Uvalde	0	—	—
Kinney	0	—	—

3. Data collection

This screening draws on a compiled data-center inventory plus authoritative geographic layers. The following table documents each data source, its content, origin, and status.

3.1 Data sources used in this screening

Dataset	Content	Source / method	Status
Data-center inventory	28 facilities: name, operator, location (lat/lon), capacity (MW), build year, status, cooling type	Operator disclosures; trade press; Texas TDLR construction filings; public facility registries (datacentermap, baxtel)	14 of 28 have capacity; remainder unquantified
Estimated water use	Acre-feet/year per facility	Computed from capacity $\times$ WUE (Section 4)	Derived
County boundaries	7 study-county polygons	US Census cartographic boundaries (public GeoJSON)	Authoritative
Edwards recharge & contributing zones	Regulatory zone polygons	TCEQ Edwards Aquifer Protection Program (30 TAC 213) / EAA	Approximate placeholder; official shapefile pending
Major springs	Comal & San Marcos locations; flow context	USGS / TWDB	Locations used; flow series illustrative
Operator WUE coefficients	Water-use efficiency (L/kWh)	Operator sustainability disclosures; industry average	Uniform 1.8 L/kWh applied (see 4.2)

3.2 Datasets planned for the full study (not used in this preliminary screening)

The companion full study incorporates additional hydrologic datasets that are not part of this preliminary screening: groundwater levels (USGS / TWDB Water Data for Texas), spring-flow records (USGS / TWDB), permitted/used pumping (TWDB), GRACE satellite storage (NASA), MODIS evapotranspiration (NASA), precipitation, drought-stage, land use / land cover, and soils / recharge-zone boundary layers. These support the AI/ML modeling and GIS vulnerability work and will refine the present estimates.

Data-quality summary. Of 28 catalogued sites: 14 quantified, 9 unquantified (capacity to be determined), 4 flagged for verification (possible mislabels), and 1 reported separately (Samsung fab). The full inventory is in Appendix A; verification findings are in Section 7.

4. Methodology and calculations

Every quantitative result in this report follows from the steps below. Constants are listed first, then the water-use model, a worked derivation, per-facility examples, and the aggregation, exposure, and projection calculations.

4.1 Constants and conversion factors

Quantity	Value	Note
Applied water-use efficiency (WUE)	1.8 L/kWh	Industry average; applied uniformly in the combined dataset
Assumed load factor	100%	Full-load assumption; scalable in the dashboard
Operating hours per year	8,760 h	24×365
Acre-foot conversion	$1 \text{ ac-ft} = 1233.48 \text{ m}^3$	Standard
Derived water factor	$12.78 \text{ ac-ft} / \text{MW} \cdot \text{yr}$	See derivation in 4.3
Household-equivalent factor	$0.4 \text{ ac-ft} / \text{household} \cdot \text{yr}$	Municipal approximation for context

4.2 Water-use model

For a facility of known capacity, estimated annual water use is:

$$\text{Water (m}^3\text{/yr)} = \text{Capacity (MW)} \times 1,000 \text{ (kW/MW)} \times \text{Load factor} \times 8,760 \text{ (h/yr)} \times \text{WUE (L/kWh)} \div 1,000 \text{ (L/m}^3\text{)}$$

$$\text{Water (ac-ft/yr)} = \text{Water (m}^3\text{/yr)} \div 1,233.48$$

Operator-specific WUE. Published operator coefficients vary widely (Microsoft 0.49, Google 1.10, Meta 0.26, Amazon 0.18 L/kWh; industry average 1.8). The combined dataset used here applies the conservative-direction industry average of 1.8 L/kWh uniformly; operator-specific coefficients are a refinement for the full study.

Operator	WUE (L/kWh)
Microsoft	0.49
Google	1.10
Meta	0.26

Operator	WUE (L/kWh)
Amazon	0.18
Industry average (applied)	1.80

4.3 Derivation of the uniform water factor

Holding load factor at 100% and WUE at 1.8 L/kWh, one megawatt yields:

- 1 MW = 1,000 kW; $\times 8,760 \text{ h/yr} = 8,760,000 \text{ kWh/yr}$.
- $\times 1.8 \text{ L/kWh} = 15,768,000 \text{ L/yr} = 15,768 \text{ m}^3/\text{yr}$.
- $\div 1,233.48 \text{ m}^3/\text{ac-ft} = 12.78 \text{ ac-ft/yr}$.

Therefore *Water (ac-ft/yr)* $\approx$ *Capacity (MW)* $\times$ **12.78**. This single factor reproduces every quantified value in the dataset.

4.4 Worked per-facility examples

Applying the model at representative capacities (100% load):

Capacity (MW)	Annual electricity (kWh)	Water (L)	Water (m ³)	Water (ac-ft/yr)
50	438,000,000	788,400,000	788,400	639.2
100	876,000,000	1,576,800,000	1,576,800	1,278.3
150	1,314,000,000	2,365,200,000	2,365,200	1,917.5
200	1,752,000,000	3,153,600,000	3,153,600	2,556.7

Example: a 100 MW facility $\rightarrow$ 876,000,000 kWh/yr $\rightarrow$ 1,576,800,000 L $\rightarrow$ 1,576,800 m³ $\rightarrow$ 1,278.3 ac-ft/yr. This matches the Microsoft Azure San Antonio entry.

4.5 Aggregation

County and operator totals are simple sums of per-facility estimates over the quantified set (the 14 sites with capacity). The quantified total is 15,928 ac-ft/yr on 1,246 MW.

4.6 Recharge-zone exposure

A facility is counted as on the recharge zone if its coordinates fall inside the recharge-zone polygon, tested by standard ray-casting (point-in-polygon). Against the current approximate boundary, 6 of 14 quantified sites are inside, totaling ~5,753 ac-ft/yr (~36% of quantified water). These counts will change when the official TCEQ/EAA boundary replaces the placeholder.

4.7 Household-equivalent context

For communication only, estimated water use is expressed in household-equivalents at 0.4 ac-ft/household·yr: $15,928 \div 0.4 \approx 39,820$ households.

4.8 Growth and historical rate

Cumulative quantified water is accumulated by facility build year from 2008 to 2027. The compound annual growth rate (CAGR) over the most recent decade is:

$$CAGR(2017-2027) = (15,928 / 4,602)^{(1/10)} - 1 = 13.2\% \text{ per year.}$$

4.9 Projection to 2050

Projected water use compounds the 2027 quantified total forward at an assumed rate r :

$$W(2050) = W(2027) \times (1 + r)^{(2050 - 2027)} = 15,928 \times (1 + r)^{23}.$$

Because the horizon is 23 years, results are extremely sensitive to r and are presented as illustrative scenarios (Section 5.4), not forecasts.

4.10 Load-factor sensitivity

Estimated water scales linearly with the load-factor assumption. Lowering the assumed load factor from 100% to 60%, for example, multiplies every estimate by 0.60, reducing the quantified total to ~9,557 ac-ft/yr. The dashboard exposes this as a slider.

5. Results

5.1 Totals (quantified sites)

Metric	Value
Quantified facilities	14 of 28
Total capacity	1,246 MW
Total estimated water use	15,928 ac-ft/yr
On recharge zone (approx.)	6 sites · ~5,753 ac-ft/yr
Household-equivalent	~39,820 households

5.2 By county

County	Sites	Capacity (MW)	Water (ac-ft/yr)	Share of water
Bexar	7	606	7,747	49%
Medina	2	350	4,474	28%
Hays	1	150	1,918	12%
Travis	4	140	1,790	11%

5.3 By operator

Operator	Sites	Water (ac-ft/yr)	Share
Microsoft	2	3,835	24%
Black Mountain	1	2,557	16%
Google	1	1,918	12%
Meta	1	1,918	12%
QTS	2	1,662	10%
Vantage	1	1,227	8%
CyrusOne	2	767	5%
Rackspace	1	639	4%
Oracle	1	639	4%
Stream Data Centers	1	511	3%
Apple	1	256	2%

5.4 Illustrative 2050 projection scenarios

From the 2027 quantified base of 15,928 ac-ft/yr, compounded to 2050:

Annual growth rate	Projected 2050 water (ac-ft/yr)	× 2027 base
5%/yr	48,923	3.1×
8%/yr	93,521	5.9×
13%/yr ($\approx$ historical)	264,829	16.6×
20%/yr	1,055,188	66.2×

These are sensitivity scenarios, not forecasts; the wide spread illustrates how strongly the 23-year horizon depends on the assumed rate.

6. Data quality, assumptions, and limitations

- Estimates, not measurements: water figures derive from capacity and a WUE coefficient, not metered withdrawals.
- Conservative floor: 14 of 28 sites lack capacity and are excluded from the total, which therefore understates the region.
- Uniform WUE: a single 1.8 L/kWh is applied; operator-specific coefficients and cooling-type detail would refine results.
- Full-load assumption: 100% utilization overstates relative to real load factors; the dashboard slider shows lower-load scenarios.
- Approximate aquifer boundary: recharge/contributing zones are placeholders; "on recharge zone" counts are provisional until the official TCEQ/EAA shapefile is substituted.

- Separated fab: Samsung Austin (process water at fab scale) is reported separately and excluded from the data-center total.
- Illustrative springflow: the Comal Springs series in the dashboard is placeholder context, not the analysis dataset.
- Projection caveat: 2050 figures are sensitivity scenarios, not forecasts.

7. Capacity research — findings on unquantified sites

A targeted review of the 14 sites lacking capacity produced data-quality corrections more than new megawatt figures. Key outcomes:

- Identities resolved: the registry-coded SAT15 and SAT40 entries are Microsoft San Antonio facilities (reassigned from “Unknown”).
- Confirmed but small: H5 San Antonio (downtown colo) and American Tower Hwy 90 (~1 MW edge node) — negligible water; exact MW not published.
- Real but mis-tagged: Iron Mountain “Austin” is a planned Williamson-County campus (outside the study area) that disclosed ~34 ac-ft/yr; its row’s county/status are incorrect.
- Likely phantom / mislabeled: Aligned Austin, T5 Austin, and Switch San Marcos do not match any discrete facility in those operators’ portfolios and should be verified or dropped.
- Undisclosed: the two AWS sites and remaining registry-coded entries have no public per-site capacity and remain blank, with a footnote, per a conservative approach.

8. Policy findings and options

The screening maps each finding to a policy lever on the table. Options are presented for deliberation, not endorsed.

Finding	Policy option	Lever
Facility water use is estimated, not reported; half of sites lack capacity	Require large data centers to report metered water use, source, and cooling to a public registry	State reporting statute / TCEQ rule
Estimated use scales with cooling choice; evaporative sites dominate	Set water-use-efficiency benchmarks or tiered standards for new large loads	Efficiency standard tied to permitting / incentives
Several sites fall on or near the recharge zone	Add recharge-zone review to siting; steer high-water cooling away from sensitive zones	EAA review · local land use
Most modeled water is consumptive (evaporated)	Require or incentivize reclaimed / non-potable water and closed-loop cooling	Reuse mandate · reclaimed-water tariff
Most capacity is recent or planned; growth can outpace planning	Tie new large-load hookups to a cumulative water-availability finding	EAA permit conditions · GCD coordination

9. References and data sources

- Texas Commission on Environmental Quality (TCEQ) — Edwards Aquifer Protection Program, 30 TAC Chapter 213 (recharge, transition, contributing zones).
- Edwards Aquifer Authority (EAA) — aquifer system and regulatory-zone mapping.
- US Census Bureau — cartographic county boundary files.
- US Geological Survey (USGS) / Texas Water Development Board (TWDB) — springs and hydrologic data.
- Texas Department of Licensing and Regulation (TDLR) — construction/registration filings used to identify facilities.
- Public data-center registries (datacentermap, baxtel) and operator sustainability disclosures — facility attributes and WUE coefficients.

Appendix A — Full facility inventory (28 sites)

All catalogued facilities with annotations. “Include” indicates treatment in this screening: Yes = quantified; Pending = capacity to be determined; Exclude = flagged for verification; Separate = fab reported separately.

Facility	Operator	County	MW	Year	Status	Cooling	Water (ac-ft/yr)	Include
Rackspace Technology HQ	Rackspace	Bexar	50	2008	Operating	Evaporative	639	Yes
Stream Data Center San Antonio I	Stream Data Centers	Bexar	40	2010	Operating	Evaporative	511	Yes
CyrusOne San Antonio	CyrusOne	Bexar	30	2012	Operating	Evaporative	384	Yes
Microsoft Azure San Antonio	Microsoft	Bexar	100	2014	Operating	Evaporative	1,278	Yes
H5 Data Centers San Antonio	H5 Data Centers	Bexar	—	2015	Operating	Air	—	Pending — capacity TBD
QTS San Antonio	QTS	Bexar	90	2018	Operating	Evaporative	1,151	Yes
Texas Research Park SAT46	Unknown (likely Microsoft)	Bexar	—	2020	Operating	Unknown	—	Pending — capacity TBD
SAT15 Data Center	Microsoft	Bexar	—	2026	Under Construction	Unknown	—	Pending — capacity TBD
SAT40 Data Center	Microsoft	Bexar	—	2026	Planned	Unknown	—	Pending — capacity TBD
American Tower Data Center Hwy 90	American Tower	Bexar	—	2026	Planned	Unknown	—	Pending — capacity TBD
Vantage Data Center San Antonio	Vantage	Bexar	96	2027	Planned	Evaporative	1,227	Yes
Black Mountain Power San Antonio	Black Mountain	Bexar	200	2027	Planned	Evaporative	2,557	Yes

Facility	Operator	County	MW	Year	Status	Cooling	Water (ac-ft/yr)	Include
EX West Green Data Center	EX West	Bexar	—	2027	Planned	Unknown	—	Pending — capacity TBD
New Braunfels Data Center	Unknown	Comal	—	2024	Planned	Unknown	—	Pending — capacity TBD
Meta Data Center Kyle	Meta	Hays	150	2021	Operating	Evaporative	1,918	Yes
Amazon AWS Data Center Buda	Amazon	Hays	—	2022	Operating	Evaporative	—	Pending — capacity TBD
Switch Data Center San Marcos	Switch	Hays	—	2023	Operating	Evaporative	—	Exclude — verify (likely phantom)
Google Data Center Castroville	Google	Medina	150	2023	Operating	Evaporative	1,918	Yes
Microsoft Data Center Castroville	Microsoft	Medina	200	2025	Under Construction	Evaporative	2,557	Yes
Amazon AWS Castroville	Amazon	Medina	—	2026	Planned	Evaporative	—	Pending — capacity TBD
Samsung Semiconductor Austin	Samsung	Travis	—	1996	Operating	Evaporative	—	Separate (fab)
Apple Data Center Austin	Apple	Travis	20	2012	Operating	Evaporative	256	Yes
Iron Mountain Austin	Iron Mountain	Travis	—	2013	Operating	Air	—	Exclude — planned/verify
CyrusOne Austin	CyrusOne	Travis	30	2014	Operating	Evaporative	384	Yes
Oracle Cloud Austin	Oracle	Travis	50	2016	Operating	Evaporative	639	Yes
QTS Austin Data Center	QTS	Travis	40	2017	Operating	Evaporative	511	Yes

Facility	Operator	County	MW	Year	Status	Cooling	Water (ac-ft/yr)	Include
T5 Data Centers Austin	T5 Data Centers	Travis	—	2019	Operating	Air	—	Exclude — verify (likely phantom)
Aligned Data Centers Austin	Aligned	Travis	—	2021	Operating	Evaporative	—	Exclude — verify (likely phantom)